

Biased Random Walk on Partitions and Hyperoctahedral Hall–Littlewood Polynomials

Adrián Vidal

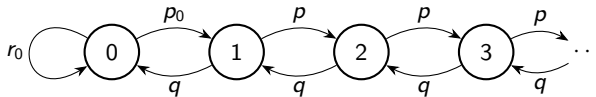
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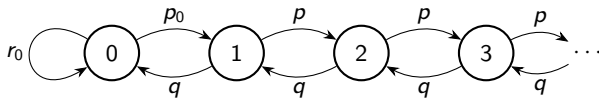
Biased random walk on $\mathbb{N}_0$

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A particle moving on $\mathbb{N}_0 = \{0, 1, 2, \dots\}$ can be modeled as a Markov chain with a **reflecting boundary** at 0.



Setting $p_0 = p$, $r_0 = q$:

$$q = \frac{b}{b + b^{-1}}, \quad p = \frac{b^{-1}}{b + b^{-1}}, \quad b \in (0, \infty).$$

Main question: compute $\mathbb{P}(X_n = j \mid X_0 = i) = \mathbf{P}_{i,j}^n$.

1-particle solution via Bernstein–Szegő polynomials

Diagonalize $\mathbf{P}$ on $\ell^2(\mathbb{N}_0)$. Eigenfunctions can be expressed in terms of **Bernstein–Szegő polynomials**:

$$Q_j(\xi) = b^j B_j^{(\alpha)}(e^{i\xi}), \quad B_j^{(\alpha)}(z) = c(z; \alpha) z^j + c(z^{-1}; \alpha) z^{-j}, \quad c(z; \alpha) = \frac{1 - \alpha z^{-1}}{1 - z^{-2}},$$

with $\alpha = b$, $z = e^{i\xi}$. Eigenvalues: $E(\xi) = \frac{2 \cos \xi}{b + b^{-1}}$ for $\xi \in [0, \pi]$ (plus $E = 1$ if $b > 1$).

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Orthogonality relation

$$\frac{1}{2\pi} \int_0^\pi B_l^{(\alpha)}(e^{i\xi}) B_k^{(\alpha)}(e^{i\xi}) \Delta(\xi; \alpha) d\xi + B_l^{(\alpha)}(\alpha) B_k^{(\alpha)}(\alpha) \Delta_\alpha = \delta_{lk},$$

where $\Delta(\xi; \alpha) = |c(e^{i\xi}; \alpha)|^{-2}$, $\Delta_\alpha = (\alpha - \alpha^{-1})/\alpha$ if $|\alpha| > 1$, else 0.

Transition probabilities as a **spectral integral**:

$$\mathbf{P}_{i,j}^m = \frac{1}{2\pi} \int_0^\pi E(\xi)^m Q_i(\xi) Q_j(\xi) \Delta(\xi; \alpha) d\xi + (\text{discrete term if } b > 1).$$

Extension to N variables: the state space $\Lambda^{(N)}$

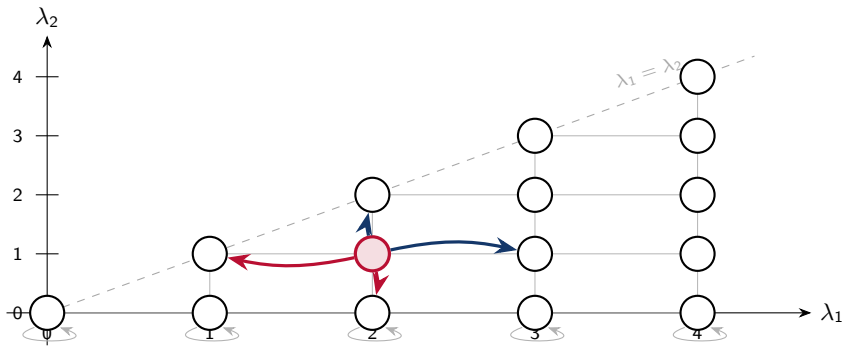
Replace $\mathbb{N}_0$ by the set of **integer partitions with at most N parts**:

$$\Lambda^{(N)} := \{ \lambda = (\lambda_1, \dots, \lambda_N) \in \mathbb{N}_0^N \mid \lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_N \geq 0 \}.$$

Physical picture: N ordered particles on $\mathbb{N}_0$ with *interaction statistics* governed by the parameter $t \in [-1, 1]$.

A picture of the chain on $\Lambda^{(2)}$

$$\Lambda^{(2)} = \{(\lambda_1, \lambda_2) \in \mathbb{N}_0^2 : \lambda_1 \geq \lambda_2 \geq 0\}.$$



The N -particle transition operator $\mathbf{P}^{(N)}$

For $\varphi \in \ell^2(\Lambda^{(N)})$, define

$$\begin{aligned} \mathbf{P}^{(N)}(\varphi)_\lambda = \frac{1}{\Sigma} & \left(\sum_{\substack{1 \leq j \leq N \\ \lambda + e_j \in \Lambda^{(N)}}} p_j(\lambda; t) \varphi_{\lambda + e_j} \right. \\ & \left. + \sum_{\substack{1 \leq j \leq N \\ \lambda - e_j \in \Lambda^{(N)}}} q_j(\lambda; t) \varphi_{\lambda - e_j} + \delta_{\lambda_N} q_N(\lambda; t) \varphi_\lambda \right), \end{aligned}$$

with normalization constant $\Sigma = b[N]_t + b^{-1}[N]_{t-1}$.

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with normalization constant $\Sigma = b[N]_t + b^{-1}[N]_{t-1}$.

For $N = 1$

Reduces to the 1-particle operator $\mathbf{P}$ (parameter t plays no role).

Connection

Coincides (up to normalization and parameter identification) with the N -particle Hamiltonian of the *semi-infinite q -boson system* [van Diejen–Emsiz, 2014].

Hall–Littlewood polynomials with hyperoctahedral symmetry

Let $\mathbf{z} = (z_1, \dots, z_N)$, $|\alpha| \leq 1$, $t \in [-1, 1]$, $\lambda \in \Lambda^{(N)}$. Define:

$$R_\lambda(\mathbf{z}; \alpha, t) := \sum_{\substack{\sigma \in S_N \\ \varepsilon \in \{-1, 1\}^N}} C(\mathbf{z}_\sigma^\varepsilon; \alpha, t) \prod_{i=1}^N z_{\sigma(i)}^{\varepsilon_i \lambda_i},$$

where the **weight function** is

$$C(\mathbf{z}; \alpha, t) = \prod_{i=1}^N c(z_i; \alpha) \cdot \prod_{1 \leq i < j \leq N} \frac{z_i - tz_j}{z_i - z_j} \cdot \frac{z_i z_j - t}{z_i z_j - 1}.$$

Special cases:

- ▶ $N = 1$: $R_\lambda \rightarrow B_n^{(\alpha)}$ (Bernstein–Szegő)
- ▶ $t = 0$: Schur-type polynomials (generalize Schur functions)

Symmetry group:

R_λ is invariant under the *hyperoctahedral group*

$$B_N = S_N \ltimes (\mathbb{Z}/2\mathbb{Z})^N$$

(signed permutations of coordinates).

Orthogonality relation

For $|\alpha| < 1$, $t \in [-1, 1]$, and $\lambda, \mu \in \Lambda^{(N)}$, the orthogonality turns the eigenfunction formula into a spectral decomposition.

$$\langle R_\lambda, R_\mu \rangle_{\Delta^{(N)}} = \mathcal{N}(\lambda) \delta_{\lambda\mu}, \quad \Delta^{(N)}(\xi) = \frac{1}{(2\pi)^N |C(e^{i\xi}; \alpha, t)|^2}.$$

The inner product is the integral over $[0, \pi]^N$.

Spectral decomposition. The **normalized eigenfunctions** ($z_j = e^{i\xi_j}$)

$$Q_\lambda^{(N)}(\mathbf{z}) = \frac{R_\lambda(z_1, \dots, z_N)}{R_\lambda(\alpha, \alpha t, \dots, \alpha t^{N-1})}$$

satisfy $\mathbf{P}^{(N)} Q_\lambda^{(N)}(\mathbf{z}) = E^{(N)}(\mathbf{z}) Q_\lambda^{(N)}(\mathbf{z})$, with eigenvalues

$$E^{(N)}(\mathbf{z}) = \frac{1}{\Sigma} \sum_{j=1}^N (z_j + z_j^{-1}).$$

Integral representation

$$\mathbf{P}_{\lambda, \mu}^{(N), m} = \int_{[0, \pi]^N} (E^{(N)}(e^{i\xi}))^m Q_{\lambda}^{(N)}(e^{i\xi}) \overline{Q_{\mu}^{(N)}(e^{i\xi})} \Delta^{(N)} d\xi_1 \cdots d\xi_N.$$

Integral representation

$$\mathbf{P}_{\lambda,\mu}^{(N),m} = \int_{[0,\pi]^N} \left(E^{(N)}(e^{i\xi})\right)^m Q_{\lambda}^{(N)}(e^{i\xi}) \overline{Q_{\mu}^{(N)}(e^{i\xi})} \Delta^{(N)} d\xi_1 \cdots d\xi_N.$$

- ▶ Long-time behavior
- ▶ Convergence to equilibrium
- ▶ Mixing rates
- ▶ Parameter deformations

-  M. Domínguez de la Iglesia, *Orthogonal Polynomials in the Spectral Analysis of Markov Processes: Birth-Death Models and Diffusion*, Cambridge University Press, 2021.
-  J. F. van Diejen and E. Emsiz, *The semi-infinite q -boson system with boundary interaction*, Letters in Mathematical Physics, 104 (2014), 103–113.
-  A. Vidal, *El espectro del laplaciano discreto en $\mathbb{Z}_+$ mediante la ortogonalidad de polinomios de Bernstein–Szegő y funciones de Schur generalizadas*, Master's thesis, Universidad de Talca, 2023.

Thank you for your attention!

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Extra: conceptual map

The whole construction follows the same spectral principle:

Markov chain $\longrightarrow$ transition operator $\longrightarrow$ orthogonal eigenfunctions $\longrightarrow$ transition probabilities.

For a discrete-time Markov chain, the m -step transition probabilities are entries of the m -th power of the transition matrix:

$$\mathbb{P}(X_m = \mu \mid X_0 = \lambda) = (\mathbf{P}^{(N)})_{\lambda, \mu}^m.$$

After diagonalization, applying $\mathbf{P}^{(N)}$ m times corresponds to raising the eigenvalue to the m -th power:

$$\mathbf{P}^{(N)} Q = E Q \implies (\mathbf{P}^{(N)})^m Q = E^m Q.$$

Key idea

The time dependence of the Markov chain is encoded in the factor $(E^{(N)})^m$.

Extra: particles, partitions and allowed moves

A partition

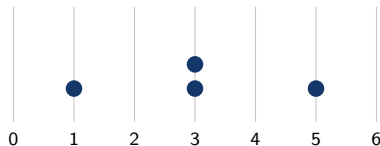
$$\lambda = (\lambda_1, \dots, \lambda_N) \in \Lambda^{(N)}$$

represents N ordered particles on $\mathbb{N}_0$.

Example:

$$\lambda = (5, 3, 3, 1).$$

This means: one particle at 5, two particles at 3, and one particle at 1.



The basic moves are

$$\lambda \mapsto \lambda + e_j, \quad \lambda \mapsto \lambda - e_j,$$

but only if the new vector still belongs to $\Lambda^{(N)}$.

Why the ordering matters

The particles cannot cross. Algebraically, this is the condition $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_N \geq 0$.

Extra: probabilities of transition in one step

Key notation (*only to write the transition probabilities compactly*):

$$\begin{aligned} m_l(\lambda) &= |\{j : \lambda_j = l\}| && \text{multiplicity of level } l \\ \vartheta_l(\lambda) &= \max\{j : \lambda_j = l\} && \text{rightmost index at level } l \\ [n]_t &= \frac{1 - t^n}{1 - t} && t\text{-analog of } n \end{aligned}$$



$$p_j(\lambda) = b^{-1} t^{-(N - \vartheta_{\lambda_j})} [m_{\lambda_j}(\lambda)]_{t^{-1}} \quad (1)$$



$$q_j(\lambda) = b t^{N - \vartheta_{\lambda_j}} [m_{\lambda_j}(\lambda)]_t \quad (2)$$

Extra: meaning of the parameters

Parameter	Interpretation
b	Controls the bias of the walk. In the one-particle model, $q = \frac{b}{b + b^{-1}}, \quad p = \frac{b^{-1}}{b + b^{-1}}.$ Thus $b > 1$ means drift toward the origin.
t	Hall–Littlewood deformation parameter. It controls the interaction between particles through the t-integers $[m]_t = \frac{1 - t^m}{1 - t}.$ For a probabilistic interpretation, one usually works in a range where all transition probabilities are nonnegative.
α	Boundary/spectral parameter appearing in $c(z; \alpha) = \frac{1 - \alpha z^{-1}}{1 - z^{-2}}.$

Extra: why hyperoctahedral symmetry?

The polynomial $R_\lambda(\mathbf{z}; \alpha, t)$ is built by summing over

$$S_N \ltimes \{\pm 1\}^N.$$

This group contains two types of symmetries:

► **Permutations of variables:**

$$z_i \longleftrightarrow z_j.$$

► **Inversions of variables:**

$$z_i \longleftrightarrow z_i^{-1}.$$

Therefore the relevant symmetry group is the hyperoctahedral group, equivalently the Weyl group of type B_N/C_N :

$$B_N = S_N \ltimes (\mathbb{Z}/2\mathbb{Z})^N.$$

Extra: special parameter regimes

- ▶ $N = 1$:

$$R_\lambda(z; \alpha, t) = B_\lambda^{(\alpha)}(z).$$

The multivariable theory reduces to Bernstein–Szegő polynomials.

- ▶ $t = 0$: The Hall–Littlewood deformation specializes to a Schur-type regime.

- ▶ $t \rightarrow 1$: Formally,

$$[m]_t = \frac{1 - t^m}{1 - t} \longrightarrow m.$$

The deformed multiplicity becomes the ordinary multiplicity.

- ▶ $b = 1$: In the one-particle model,

$$p = q = \frac{1}{2}.$$

The walk is unbiased.